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List of tasks today	time	advance preparation	Date
Glucose solution prepare	30 mins	done	1 Jun
syringe filtering	30 mins	done	
Make PDMS (7.5%)	2 hours	done	
Make fresh GUV lipid solution	30 mins	done	
DoTAP SUV lipid solution in chloroform preparation	30 mins	done	
Deposit SUV solution in MCT for drying		done	
Deposit GUV solution on ITO for drying		done	
BSA coating pipettes dip	10 mins	day 4	
Electroform GUVs - DOPS-PC-Chol	1 hour	done	2 Jun
pressure sensor bubble remove and glucose medium set	2 hours	done	
New pipette couple with pressure sensor using glucose media	30 mins	done	
Evaluate quality/ stability of GUV via aspiration		done	
Calibrate pressure sensor's zero point	1 hour	done	
Setup the flow control line with Nemesys syringe pump	1 hours	day 4	
Do fusion with aspiration experiment		day 5	
cut 4 sample holders (PDMS)		done	3 Jun
Sample holder holes for low flow rate setup	30 mins	done	
Plasma cleaning and sticking glass with PDMS		done	
Pull more pipettes	3 hours	done	
Make pipettes long and short so that they are horizontal		done	
forge pipettes	2 hours	done	
Make Fresh SUVs with membrane		day 4	
Dilute the SUVs	10 mins	day 4	
Data Analysis 1	3 hrs	done	
Need to have better control over concentration of SUV		day 5	
Write the experiment protocol in today.txt	1 hour	day 5	
clip the tube to stop flow when not in use		day 5	
Change the concentration and repeat the experiment		day 5	
Change the charges and repeat the experiment		day 5	
Added PVA to water line	30 mins	day 5	
BSA solution prepare	30 mins	later	
Iisc SUV characterization book again		Later	
data analysis 2		later	

```

import numpy as np
print("Hello,world")

### Let's make 90% charged DOTAP-DOPC (Positive, non-fluorescent)
Chemicals = ['DOPS','DOPC','LissRhod-PE', 'Cholesterol', 'NBD-DOTAP']
Concentrations = [2.5,2.5,0.05,2.5,0.01]; # In chloroform
MixtureVolumes = [20,150,0,0,0]; # in microliters
Molecular_Weights = [809.518,785.593,1300.712,386.65,710.34]

# Number of Weights in mixture
Weights = np.multiply(MixtureVolumes, Concentrations)
# Percentage of moles in mixture
WeightPercentages = np.array(Weights) / np.sum(Weights) * 100

Moles = np.array(Weights)/np.array(Molecular_Weights)
MolePercentages = np.array(Moles) / np.sum(Moles) *100
np.set_printoptions(suppress=True, precision=5)

print('Moles')
print(Moles)
print('Weights')
print(Weights)
# Percentage of moles in mixture
print('weight Percentages')
print(np.round(WeightPercentages,2))
print('Mole Percentages')
print(np.round(MolePercentages,2))

```